MTH1001-A

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

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Mathematical Structures

Module Leader: Dr Gihan Marasingha

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) (i) Give the parse tree for the expression $2 \times 3 = 1 + 6 \times 7$.

(ii) State the substitution principle.

(iii) Let a and b be real numbers. Suppose P is the predicate on $\mathbb{R}$ given by $P(x) \coloneqq 5 + a^2x \leq bx^2 + (a+3)$. Suppose $h_1 : a + 3 = 2b$ and $h_2 : 5 + a^2(a+3) \leq b(a+3)^2 + (a+3)$. What is the result of substituting h_1 into h_2 with predicate P? (10)

(b) In this question, select **one or more** possible deductions from a list of premises. You need not provide reasoning.

(i) Given the premises $p \wedge q$ and p, which of the following can be deduced: p, q, $p \wedge q$, $p \vee q$, $\neg p$, $\neg q$?

(ii) Given the premises $p \vee q$ and q, which of the following can be deduced: p, q, $p \wedge q$, $p \vee q$, $\neg p$, $\neg q$?

(iii) Given the premises $p \to q$ and q, which of the following can be deduced: p, q, $p \wedge q$, $p \vee q$, $\neg p$, $\neg q$?

(iv) Given the premises $p \to q$ and $\neg q$, which of the following can be deduced: p, q, $p \wedge q$, $p \vee q$, $\neg p$, $\neg q$? (8)

(c) (i) Let A and B be types. Let $f : A \to B$ be a function. Give the definition of 'f is injective'.

(ii) Which of the following functions are injective? In each case give a proof of your answer. You may use all familiar properties of arithmetic.

$f : \mathbb{Z} \to \mathbb{R}$, where $f(x) = (x - \frac{1}{3})^2$,
$g : \mathbb{Z} \to \mathbb{R}$, where $g(y) = (y - \frac{1}{2})^2$. (10)

(d) Let $A = \{5, 7, 8\}$, $B = \{2, 3, 9\}$, $C = \{5, 3, 9\}$, $D = \{7, 8\}$ be sets on the type $\mathbb{N}$. Write each of the following sets as lists of elements enclosed in braces **and** state the number of elements in each set. You need not provide reasoning.

(i) $A \cap D$, (ii) $B \cup C$, (iii) $A \setminus D$,
(iv) $\mathcal{P}(A \setminus D)$, (v) $\mathcal{P}(\mathcal{P}(A \setminus D))$, (vi) $\mathcal{P}(A \setminus D) \times D$. (12)

(e) Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers. Let a be a real number.

(i) Define what it means for $(a_n)_{n=0}^{\infty}$ to converge to a.

(ii) Suppose $(a_n)_{n=0}^{\infty}$ converges to a. Let $f : \mathbb{N} \to \mathbb{N}$ be the function given by $f(n) \coloneqq n$, if n is odd and $f(n) \coloneqq n/2$, if n is even. Using only the definition of convergence (and all familiar properties of arithmetic), prove that the sequence $(a_{f(n)})_{n=0}^{\infty}$ converges. (10)

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SECTION B

2. (a) Give a Fitch-style proof of $((p \leftrightarrow q) \wedge \neg q) \rightarrow \neg q$, for propositions p and q. Your proof should use only the rules of inference. (12)

   (b) Let S be a set of real numbers.

   (i) Give the definition of an *upper bound* of S.

   (ii) Give the definition of a *lower bound* of S.

   (iii) Given that a is a lower bound of S and b is an upper bound of S, does it follow that $a \leq b$? Justify your answer.

   (iv) Give the definition of a *supremum* of S.

   (v) Given that a is a lower bound of S and c is a supremum of S, does it follow that $a \leq c$? Justify your answer. (13)

   [25]

3. (a) Let A and B be types, let S and T be sets of elements of type A and let $f : A \rightarrow B$ be a function.

   (i) Give the definition of the image $f(S)$.

   (ii) Given that f is injective and given that $f(S) = f(T)$, does it follow that $S = T$? Justify your answer. (10)

   (b) A binary operator $\odot$ is defined on $\mathbb{N}$ recursively by the rules:

   - for every $m : \mathbb{N}$, $m \odot \mathsf{O} \coloneqq m$,
   - for all $m, n : \mathbb{N}$, $m \odot \mathsf{S}(n) \coloneqq \frac{m}{n+1}(m \odot n) + m$.

   (i) For each $n : \mathbb{N}$, prove by induction that $m \odot n$ is a polynomial in the variable m with rational coefficients, degree $n + 1$, and constant term zero. You may freely use all results of rational arithmetic.

   (ii) For each natural number $n \geq 1$, guess formulae, in terms of n, for the coefficients of the m and m^2 terms in the polynomial $m \odot n$. Prove your formulae by induction. You may freely use all results of rational arithmetic. (15)

   [25]

4. (a) Let A, B, and C be types. Let $f_1 : A \to B$ and $f_2 : A \to C$ be bijections. Define $f : A \to B \times C$ by $f(x) \coloneqq (f_1(x), f_2(x))$. Is f necessarily injective? Is f necessarily surjective? Justify your answer. (12)

   (b) Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers.

      (i) What does it mean for $\sum_{n=0}^{\infty} a_n$ to converge?

      (ii) Suppose $(a_n)_{n=0}^{\infty}$ is a sequence of non-negative real numbers. Given that $\sum_{n=0}^{\infty} a_n$ converges, does it follow that $\sum_{n=0}^{\infty} a_n^2$ converges? Justify your answer. You may freely use any result on sequences or series stated in lectures or seen in example sheets. (13)

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